THE COMPLETE GUIDE TO

RAG EVOLUTION

From Traditional RAG to Graph RAG to Agentic RAG
— Smarter Retrieval. Deeper Reasoning. Real-World Impact.



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Explore all 4 essential slides

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RAG EVOLUTION CHEAT SHEET

From Traditional RAG → Graph RAG → Agentic RAG

WHAT IS RAG?

Retrieval-Augmented Generation (RAG) combines:

External Knowledge + LLM Reasoning

Instead of relying only on model training data, the LLM retrieves relevant information from your data sources before generating a response.

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COMMON RAG USE CASES

- Enterprise Knowledge Base
- Customer Support Assistant
- Internal Company Search
- Legal Document Analysis
- Healthcare Knowledge Retrieval
- Financial Research Assistant
- Product Documentation Chatbot

WHY RAG?

Without RAG:

- Hallucinations
- Outdated Information
- No Access To Private Data
- Expensive Fine-Tuning

With RAG:

- Grounded Responses
- Real-Time Knowledge
- Domain-Specific Answers
- Lower Cost

THE EVOLUTION OF RAG

Type	Best For	Complexity
Traditional RAG	Document Search	Low
Graph RAG	Relationship Discovery	Medium
Agentic RAG	Multi-Step Reasoning	High

TRADITIONAL RAG

Think:

"Find relevant chunks and answer"

Example:

User: "What is our leave policy?"

↓

Retrieve HR document

↓

Generate answer

Best For:

- FAQs
- Internal Search
- Documentation Assistants

GRAPH RAG

Think:

"Understand relationships"

Example:

User: "Which customers are connected to Project X through Partner Y?"

↓

Traverse graph relationships

↓

Gather connected entities

↓

Generate answer

Best For:

- Knowledge Graphs
- Research
- Fraud Detection
- Complex Relationships

AGENTIC RAG

Think:

"Plan → Retrieve → Reason → Act"

Example:

User: "Compare AWS costs from last quarter and suggest optimization opportunities."

↓

Agent plans tasks

↓

Queries databases

↓

Retrieves reports

↓

Analyzes results

↓

Generates recommendations

Best For:

- AI Copilots
- Research Agents
- Autonomous Workflows
- Multi-Step Tasks

DECISION GUIDE

USE TRADITIONAL RAG IF:

- Search is enough
- Documents are independent
- Fast implementation required

USE GRAPH RAG IF:

- Relationships matter
- Multi-hop reasoning needed
- Connected knowledge exists

USE AGENTIC RAG IF:

- Complex workflows exist
- Multiple tools required
- Planning and reasoning needed

QUICK RULE OF THUMB

Search → Traditional RAG → Relationships → Graph RAG → Reason + Act → Agentic RAG

TRADITIONAL RAG DEEP DIVE

How Traditional RAG Works in Enterprise Environments

REAL-WORLD SCENARIO

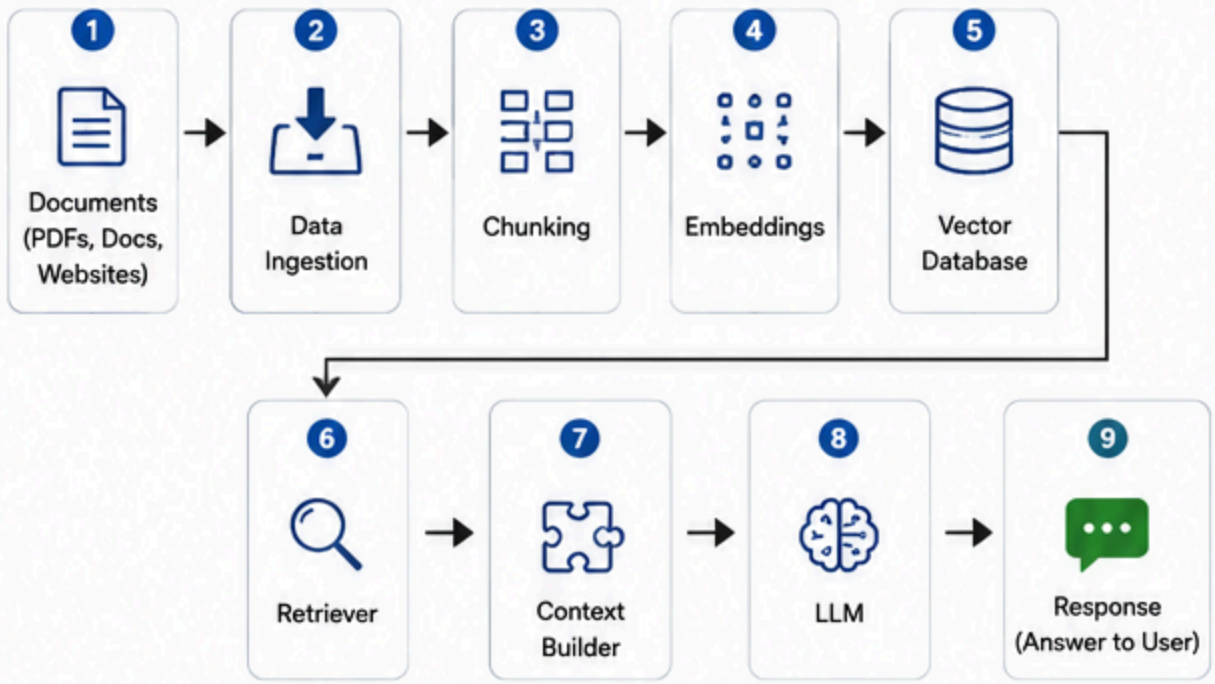
Enterprise HR Assistant

Employee asks:

"How many sick leaves can I take annually?"

System retrieves HR policy and answers.

ENTERPRISE ARCHITECTURE



COMPONENTS

1. DATA SOURCES

- PDFs
- SharePoint
- Confluence
- Websites
- Internal Documents

2. CHUNKING

Goal:

Split large documents into meaningful sections.

Popular Methods:

- Fixed Chunking
- Semantic Chunking
- Recursive Chunking
- Parent-Child Chunking

3. EMBEDDING MODELS

Convert text into vectors.

Examples:

- OpenAI Embeddings
- BGE
- E5
- Voyage AI

4. VECTOR DATABASE

Stores embeddings.

Examples:

- Pinecone
- Weaviate
- Qdrant
- Chroma
- Milvus

5. RETRIEVAL

Techniques:

- Similarity Search
- Hybrid Search
- Metadata Filtering
- Query Expansion
- Re-ranking

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PRODUCTION BEST PRACTICES

- Metadata Filtering
- Hybrid Search
- Re-Ranking Layer
- Query Rewriting
- Caching
- Evaluation Framework
- Hallucination Detection



ADVANTAGES

- Simple Architecture
- Fast Retrieval
- Easy To Scale
- Low Cost

LIMITATIONS

- Poor Multi-Hop Reasoning
- Weak Relationship Understanding
- Chunk Dependency Issues
- Context Window Constraints

IDEAL USE CASES

- Employee Assistants
- Documentation Search
- Customer Support Bots
- Product Knowledge Systems

KEY TAKEAWAY

Traditional RAG is the most common and easiest way to augment LLMs with external knowledge. It works best when your data is document-based and you need accurate, up-to-date answers with simple retrieval.

GRAPH RAG DEEP DIVE

Leveraging Relationships for Smarter Retrieval and Reasoning

REAL-WORLD SCENARIO

Financial Fraud Investigation

Question:

"Show all suspicious transactions linked to Company A through vendors and intermediaries."

Traditional RAG struggles.

Graph RAG excels.

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ENTERPRISE ARCHITECTURE

- Documents
- Entity Extraction
- Relationship Extraction
- Knowledge Graph
- Graph Database
- Graph Retrieval
- Context Generation
- LLM
- Answer

GRAPH STRUCTURE

Example:

- **Customer**
- *Owns*
- **Company**
- *Partnered With*
- **Vendor**
- *Transferred To*
- **Bank Account**

Relationships become searchable.

CORE COMPONENTS

1. ENTITY EXTRACTION

Identify:

- People
- Companies
- Products
- Events
- Locations



2. RELATIONSHIP EXTRACTION

Identify connections.

Examples:

- Works At
- Owns
- Reports To
- Purchased
- Connected With



3. GRAPH DATABASE

Examples:

- Neo4j
- TigerGraph
- Amazon Neptune
- ArangoDB



4. GRAPH RETRIEVAL

Retrieves:

Not only documents
But connected entities + relationships



PRODUCTION TIPS

- Entity Resolution
- Graph Updates
- Relationship Validation
- Graph Traversal Limits
- Hybrid Graph + Vector Search
- Graph Quality Monitoring



ADVANTAGES

- Multi-Hop Reasoning
- Relationship Awareness
- Better Explainability
- Knowledge Discovery



LIMITATIONS

- Higher Complexity
- Graph Maintenance
- Slower Setup
- Entity Extraction Challenges



IDEAL USE CASES

- Financial Intelligence
- Supply Chain Analysis
- Research Platforms
- Legal Investigation
- Fraud Detection
- Enterprise Knowledge Graphs

KEY TAKEAWAY

Graph RAG leverages entity relationships and multi-hop connections to deliver deeper insights that traditional retrieval cannot. It is ideal when relationships and context matter.

AGENTIC RAG DEEP DIVE

Autonomous Agents that Plan, Retrieve, Reason and Act

REAL-WORLD SCENARIO

AI Business Analyst

Question:

"Analyze Q1 revenue decline, compare with competitors, and recommend actions."

Agent does not simply retrieve.

It thinks and executes.

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ENTERPRISE ARCHITECTURE

1. User Query
2. Planner Agent
3. Task Breakdown
4. Retriever Agent
5. Tool Calling
6. Database Agent
7. Web Agent
8. Reasoning Agent
9. Response Generator

AGENT WORKFLOW

User Query:

"Find why sales dropped."

Agent:

1. Understand goal
2. Create plan
3. Retrieve internal reports
4. Query SQL database
5. Search competitor data
6. Compare results
7. Generate insights
8. Suggest actions

CORE COMPONENTS

1. PLANNER

Creates execution strategy.

2. MEMORY

Stores conversation state.

3. TOOL CALLING

Connects to:

- APIs
- Databases
- Search Systems
- CRMs
- External Services

4. MULTI-AGENT COLLABORATION

Examples:

Research Agent → SQL Agent → Analytics Agent → Report Agent

5. REASONING AGENT
(implicit in workflow)

Performs reasoning and decision making across retrieved information.

PRODUCTION BEST PRACTICES

- Human Approval Loops
- Tool Permission Controls
- Cost Monitoring
- Agent Observability
- Memory Management
- Fallback Mechanisms
- Guardrails

ADVANTAGES

- Complex Problem Solving
- Multi-Step Reasoning
- Autonomous Execution
- Dynamic Workflows

LIMITATIONS

- Expensive
- Longer Latency
- Hard Debugging
- Agent Drift Risks

IDEAL USE CASES

- AI Copilots
- Autonomous Analysts
- Research Agents
- Operations Automation
- Executive Assistants
- Enterprise Agent Platforms

FINAL COMPARISON

Feature	Traditional RAG	Graph RAG	Agentic RAG
Retrieval	Vector Search	Graph Traversal	Dynamic Retrieval
Reasoning	Low	Medium	High
Complexity	Low	Medium	High
Cost	Low	Medium	High
Multi-Hop Queries	Poor	Excellent	Excellent
Tool Usage	No	Limited	Extensive
Explainability	Medium	High	Medium
Enterprise Readiness	High	High	Emerging
Best For	Search	Relationships	Autonomous Tasks

RULE OF THUMB

SEARCH → Traditional RAG → **RELATIONSHIPS** → Graph RAG → **REASON + ACT** → Agentic RAG